

The Two Times of the Universe

Deriving the Radiation–Matter Dimensional Separation from Newton and Einstein

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Abstract

This paper derives the conclusion that the universe necessarily contains two ontologically distinct dimensional domains: a matter domain governed by relative time ($\tau > 0$) and a radiation domain governed by absolute time ($\tau = 0$). The derivation uses only two established sources: Newton's explicit textual distinction between absolute time and relative (clock) time in the *Principia*, and special relativity's own result that the proper time of light is identically zero while the proper time of all massive matter is strictly positive. The invariance of the speed of light, the axiom on which relativity rests, is the physical signature of Newton's absolute time: a quantity identical for all observers is not a relative quantity of the E4D (Einstein's 4D) manifold, but an absolute constant imposed upon it. The separation is formalised through a Temporal Domain Separation axiom, a temporal classification operator Π_τ , and a proof that no Lorentz transformation maps the timelike sector into the null sector. The standard interpretation of Minkowski spacetime's unification of null and timelike worldlines is rejected as a category error—a geometrical inclusion misread as an ontological inclusion: the divergence of the Lorentz factor at $v = c$, the infinite-energy barrier separating the two invariant momentum classes, and the century-long failure to unify quantum field theory with general relativity are three independent symptoms of the same structural fracture. This paper identifies the two domains—radiation dimensions ($S^6D-S^5D-S^4D$) and matter dimensions ($S^3D-S^2D-S^1D$)—as projections of the six-dimensional source S^6D . Falsification conditions are stated.

Keywords: absolute time; relative time; proper time; Newton; Minkowski spacetime; radiation–matter separation; dimensional ontology; six-dimensional cascade; SDMC

I. Introduction: The Question Relativity Never Answered

Special relativity is built upon an axiom it never explains: the speed of light in vacuum is identical for every inertial observer. Every other velocity in physics is relative—dependent on the reference frame of the measurer. Light's velocity alone is exempt. The standard resolution is to declare c a geometric property of spacetime itself. This paper argues that this resolution names the phenomenon without explaining it, and that the correct explanation was written down 218 years before relativity existed: in Newton's *Principia*, in the definition of absolute time.

The argument of this paper is a derivation, not a hypothesis. Its two premises are taken directly from the primary literature of classical and relativistic mechanics. Its conclusion—that radiation and matter occupy two geometrically irreconcilable domains—follows from those premises by elimination of the only proposed alternative, the Minkowski unification, which is shown to be structurally defective on three independent grounds.

The proposed separation is not a modification of the Lorentz transformations, of Maxwell's equations [1], or of quantum field dynamics within their established domains. It is the identification of what these structures have contained all along: two distinct invariant temporal sectors that no equation of the formalism connects.

II. Premise One: Newton's Two Times

In the Scholium to the Definitions of the *Principia* (1687) [2,3], Newton states: "Absolute, true, and mathematical time, of itself, and from its own nature, flows equably without relation to anything external [3]." He immediately contrasts this with "relative, apparent, and common time," which is "some sensible and external measure of duration by means of motion"—hours, days, months, years, the readings of clocks and the rotation of the Earth.

Two textual facts are decisive and are almost universally misrepresented in the secondary literature [4]. First, Newton explicitly assigns all humanly measurable time—every clock, every astronomical cycle—to the category of relative time. The clock time that relativity would later show to dilate and contract was, for Newton, never absolute time in the first place. Second, the phrase "flows equably" (*aequabiliter fluit*) does not describe a moving river of time; it describes a state of uniform mathematical distribution—a background condition identical at every point of space, dependent on nothing external, unaffected by any motion of matter. Newton's absolute time is not a process. It is a state.

Newton therefore bequeathed to physics a strict two-category ontology of time: an absolute background state, independent of all matter and all measurement; and a relative, measured time,

belonging to matter, clocks, and observers. The question his successors never asked is [5]: what physical entity, if any, instantiates the first category?

III. Premise Two: Einstein's Two Proper Times

Special relativity [6] delivers, as a theorem of its own four-dimensional geometry, a result that answers Newton's open question. Along any worldline, the proper time interval satisfies $d\tau^2 = dt^2 - (dx^2 + dy^2 + dz^2)/c^2$ [7]. For any massive body, which must travel at $v < c$, $d\tau > 0$: its clock runs, its cells age, its processes unfold. For light, which travels along null worldlines with $v = c$, $d\tau = 0$ identically: between emission and absorption, whatever the coordinate distance and coordinate duration, the photon's own elapsed time is exactly zero.

The physical content of $\tau = 0$ must be stated without euphemism. Light does not experience a very small amount of time. It experiences no time. It has no duration, no process, no ageing, no before and no after. Its state is not "fast"; it is atemporal. A photon emitted at the recombination epoch and absorbed today in a detector has, in its own terms, an elapsed existence of zero seconds across 13.8 billion years of coordinate time.

III.1 The Temporal Classification Operator and the Separation Axiom

The bifurcation just described is not a difference of degree but a difference of temporal type, and it admits exact formalisation.

Definition 1 (Temporal Classification Operator). Let γ denote any physical worldline in Lorentzian spacetime, and let $T[\gamma] = \int \gamma d\tau$ denote the proper-time functional along γ . The temporal classification operator Π_τ is defined as:

$\Pi_\tau(\gamma) = 1$ if $T[\gamma] > 0$ (timelike: intrinsic temporal parameter exists)

$\Pi_\tau(\gamma) = 0$ if $T[\gamma] = 0$ (null: no intrinsic temporal parameter exists)

Definition 2 (Temporal Domains). The matter domain D_M and radiation domain D_R are defined as:

$D_M = \{ \gamma \mid d\tau(\gamma) > 0 \}$ — the matter domain, the domain of relative time

$D_R = \{ \gamma \mid d\tau(\gamma) = 0 \}$ — the radiation domain, the domain of absolute time

with $D_M \cap D_R = \emptyset$.

The classification is Lorentz invariant: $\Pi_\tau(\gamma)$ is identical for every observer, because proper time is an invariant of the geometry. No coordinate choice, no boost, no relabelling changes a worldline's temporal class.

Axiom 1 (Temporal Domain Separation Principle).

A physical domain possesses a single temporal ontology only if one temporal parameter serves as the intrinsic evolution parameter of every entity in the domain. In Lorentzian geometry there exist two non-interchangeable temporal structures: timelike worldlines possess a non-vanishing intrinsic proper-time parameter ($\Pi_\tau = 1$), and null worldlines possess none ($\Pi_\tau = 0$). No single temporal ontology can serve simultaneously as the intrinsic parameter of both classes. Therefore, entities of the two classes belong to two distinct temporal domains, D_M and D_R , and the separation is ontological: no continuous Lorentz transformation and no finite-energy physical process maps one class into the other (proved in Sections V.1 and V.2).

The axiom does not assert that photons lack a mathematical representation in Minkowski spacetime—they manifestly have one. It asserts that mathematical representation within a formalism and ontological membership in a temporal domain are two different relations, a distinction made precise in Section IV.1. The popular dichotomy follows as a corollary: either time shares the invariance of c —in which case $T = 0$ universally and relative time does not exist—or the invariant c and the variable clock time are structures of two different domains. There is no third option in which a single temporal ontology contains both.

Set the two premises side by side. Newton defined an absolute time: a state independent of all external things, identical everywhere, unaffected by motion. Relativity proved that light possesses a temporal condition that exactly corresponds to Newton's definition of absolute time: its temporal condition ($\tau = 0$) is independent of every reference frame, unaffected by any motion of any observer, and invariant across the entire universe. Matter, meanwhile, occupies Newton's relative time exactly: clock time, frame-dependent, dilating and contracting with motion. Relativity did not abolish Newton's two-category ontology. Relativity is the first physical theory to have populated both of Newton's categories with concrete physical referents. Radiation is not absolute time itself, nor is it the absolute-time domain itself. Radiation is content residing within the absolute-time domain, exactly as tables and chairs are content residing within a room without being the room. Absolute time is the temporal state of that domain; radiation is what exists in that state. What E4D observers detect as light is the projection of that domain's content into the matter domain—the resident of one domain, imaged upon the interior of another. Correspondingly, relative time is the temporal state of the matter domain, and massive bodies are its residents.

IV. The Invariance of c as the Signature of Absolute Time

The identification above resolves the axiom relativity never explained. Within the relativistic four-dimensional manifold—hereafter E4D, the relativistic spacetime of matter—every observer-dependent quantity transforms between frames: length, coordinate duration, simultaneity, energy components, momentum components. The manifold's invariants—the interval, the four-momentum norm—are precisely the quantities constructed to survive that transformation, and every one of them is built from

c. The constant c is therefore not one invariant among others; it is the generator of the invariant structure itself, the fixed conversion standard from which all other invariants are assembled. A constant that defines the invariant structure of a manifold is not an internal product of that manifold's dynamics. It is the external condition imposed upon it, to which its entire structure conforms.

The claim admits exact expression in the metric itself. The Minkowski line element

$$ds^2 = c^2 dt^2 - dx^2 - dy^2 - dz^2$$

partitions all worldlines into the two temporal classes of Section III.1:

$$ds^2 > 0 \text{ for } v < c \text{ (timelike, D_M)} \quad ds^2 = 0 \text{ for } v = c \text{ (null, D_R)}$$

and along any trajectory approaching the boundary,

$$\lim(v \rightarrow c^-) d\tau/dt = \lim(v \rightarrow c^-) \sqrt{(1 - v^2/c^2)} = 0.$$

The vanishing of $d\tau/dt$ at the boundary is not a coordinate effect: $d\tau$ is an invariant, and its collapse to zero is the collapse of the time parameter itself, visible identically from every frame.

The present argument does not deny that Lorentz symmetry mathematically incorporates c -invariance; it asks whether c -invariance should be interpreted as an emergent internal property of the matter domain or as a fundamental constraint defining that domain. The answer is decided by the direction of accommodation. The speed of light is precisely the frame-independent quantity described above. Its invariance is not a property that E4D generates; it is the boundary condition to which E4D is forced to conform—and the celebrated relativistic effects, time dilation and length contraction, are exactly the deformations that the matter domain must undergo in order to conform to it. Matter's space and time stretch and compress so that a constant not belonging to them remains constant. This is the behaviour of a projection surface accommodating its source, not of a container accommodating its contents.

The conclusion is direct: the invariance of c is the empirical signature, measurable from within the matter domain, of the absolute-time domain's existence. Newton's absolute time is not a metaphysical postulate. It is measured every time the speed of light is measured [8], and it has returned the same value in every frame for over a century. The two hundred years of Newtonian physics were not overturned by relativity; relativity supplied the experimental proof of Newton's deepest definition.

IV.1 Geometrical Inclusion versus Ontological Inclusion

The entire dispute between the present framework and the standard reading of Minkowski spacetime reduces to the conflation of two distinct relations. The statement

$$X \in M_4$$

asserts only that the entity X possesses a representation within the four-dimensional mathematical manifold: coordinates can be assigned to it, its four-momentum can be written, its field can be defined on the manifold. The statement

$$X \in D_i$$

asserts membership in a physical temporal domain—that the entity's mode of temporal existence is that of the domain. These are not the same relation:

Representation $\neq$ Ontology.

A map of two countries is a single sheet of paper; the single sheet does not make the two countries one territory. Minkowski geometry is the single sheet on which both temporal domains are drawn—a representational unification of genuine power and elegance. The error of the standard reading is to promote the unity of the sheet into the unity of the territory: to conclude, from the fact that null and timelike worldlines are drawn in one geometry, that radiation and matter share one temporal ontology. Sections III.1 and V demonstrate that they do not: the two classes are disjoint, invariantly distinguished, dynamically unconnected, and temporally incommensurable. Minkowski spacetime provides geometrical inclusion of both domains. It provides ontological inclusion of neither claim about their unity. Every result of this paper is compatible with the full mathematical apparatus of special relativity, because the paper's target is not the apparatus but the ontological misreading of it.

V. The Standard Misreading of Minkowski and Its Three Fractures

A precise reading of Minkowski's own words reveals that his mathematical framework supports the two-domain conclusion rather than contradicting it. As Minkowski stated at the 80th Assembly of German Natural Scientists and Physicians in September 1908: "Henceforth space by itself, and time by itself, are doomed to fade away into mere shadows, and only a kind of union of the two will preserve an independent reality [9]." His claim entails clearly that: 1) space and time cannot exist independently—every temporal structure necessarily binds to a corresponding spatial structure; and 2) the temporal structure of light ($\tau = 0$) and the temporal structure of matter ($\tau > 0$) are formally non-equivalent and irreconcilably distinct [9], which means that if space and time are inseparable—if every temporal structure necessarily binds to a corresponding spatial structure—then two irreconcilably distinct temporal structures ($\tau > 0$ and $\tau = 0$) cannot share one spatial domain.

The standard interpretation of Minkowski spacetime [7], which holds that null worldlines ($\tau = 0$) and timelike worldlines ($\tau > 0$) coexist within a single four-dimensional pseudo-Riemannian manifold as different classes of curve within one geometry, is therefore a misreading that runs against the logic of his own framework. This section demonstrates that this standard ontological reading conceals a physical fracture, on three independent grounds. In the vocabulary of Section IV.1: the unification is geometrical, and each of the three fractures below is a point at which the ontological reading of that geometry breaks.

V.1 The Lorentz Divergence and the Limit of Coordinate Smoothing

The Lorentz factor $\gamma = 1/\sqrt{1 - v^2/c^2}$ [10] diverges as $v \rightarrow c$. The standard rejoinder is that this divergence is a coordinate artefact: transforming to null coordinates ($u = ct - x$, $v = ct + x$) renders the metric smooth across the light cone, and the divergence disappears from the formalism [11]. The rejoinder must be answered directly, and it can be. Coordinate transformation is a relabelling of the same underlying physical facts; it can remove a singularity that was an artefact of labelling, but it cannot alter a single physical invariant. The relevant invariants here are two. First, the proper time along every

null worldline remains $\tau = 0$ in null coordinates, in standard coordinates, and in every coordinate system that can ever be constructed: the temporal collapse of the radiation state is coordinate-independent and is therefore a fact about the geometry, not about the map. Second, no continuous physical process transforms any timelike worldline into a null worldline: the boundary is dynamically impassable (Section V.2).

The impossibility is not merely dynamical but group-theoretical, and this is the strongest form of the argument. Define the proper-time functional over any worldline γ :

$$T[\gamma] = \int \gamma \, d\tau.$$

Proposition 1 (Lorentz Group Separation). No element L of the Lorentz group maps any timelike worldline into a null worldline.

Proof. Lorentz transformations preserve the spacetime interval and therefore preserve the causal character of every worldline. For any $\gamma_M \in D_M$, $T[\gamma_M] > 0$; for any $\gamma_R \in D_R$, $T[\gamma_R] = 0$. Since the invariant proper-time class is preserved under L , $L(\gamma_M) \notin D_R$. Therefore no Lorentz transformation connects the two sectors.

Corollary 1 (Separation of Temporal Structures). The two classes possess non-equivalent temporal structures: D_M contains worldlines with an intrinsic proper-time parameter, whereas D_R contains worldlines with no such parameter. Under Axiom 1, these two structures correspond to distinct temporal domains.

The two sectors are not connected by any element of the symmetry group of the theory. This is categorically stronger than an energy argument: it is not that crossing the boundary is expensive, but that the transformation required to cross it is absent from the group. The Lorentz group—the very structure that defines Minkowski spacetime—itself partitions the worldlines into disconnected temporal classes and contains no operation joining them.

A boundary at which an invariant physical quantity (proper time) discontinuously collapses from strictly positive to identically zero, which no physical trajectory can cross, and which no element of the theory's own symmetry group bridges, is not an interior region of a single physical domain rendered awkward by poor coordinates. It is the edge of the domain. Null coordinates smooth the mathematics of the map; they do not repair the physics of the territory. The distinction between coordinate singularity and structural boundary is decided by invariants, and every invariant testifies that $v = c$ is structural.

V.2 The Infinite-Energy Barrier and the Two Invariant Momentum Classes

Within E4D, accelerating any massive body toward c requires energy $E = \gamma mc^2$, which diverges without bound. No finite process, no finite energy, no physical mechanism of any kind carries a single particle of matter across the boundary into the radiation state. Conversely, no photon can be decelerated into rest; a photon at rest is not a slow photon but a contradiction in terms, since the massless dispersion relation admits no rest frame.

The separation is written into the four-momentum invariant itself. For any physical entity,

$$p^\mu p_\mu = m^2 c^2 \text{ [12].}$$

For every massive entity, $p^\mu p_\mu > 0$: the entity belongs to the positive invariant class, possesses a rest frame, and carries an intrinsic proper-time parameter. For every photon, $p^\mu p_\mu = 0$: the entity belongs to the null invariant class, possesses no rest frame, and carries no proper-time parameter. These are two invariant classes of the momentum space, not two values of a continuously connected variable. The mathematical limit $m \rightarrow 0$ does not carry the one class into the other:

$$\lim(m \rightarrow 0) D_M \neq D_R.$$

The argument therefore does not rely on the limit $m \rightarrow 0$. The distinction is between the invariant structures ($m > 0$, $\tau > 0$, $p^2 > 0$) and ($m = 0$, $\tau = 0$, $p^2 = 0$)—two disjoint sectors of the theory's own invariant classification—not between two numerical endpoints of one state space. A massless state is not the destination of an ever-lighter massive state; it is a different kind of state, reached by no trajectory.

Between the two classes there is no path—not a difficult path, not an expensive path, but no path. Two states between which the laws of physics themselves permit no continuous transformation are not two regions of one domain. They are two domains. The comparison with other impassable-seeming boundaries fails: the sound barrier is crossed with finite energy; phase transitions are crossed with finite energy; every boundary internal to the matter domain is finite. Only this boundary is infinite, because it is not internal.

V.3 The Forbidden Mapping and the Century of Failed Unification

If the standard reading of Minkowski spacetime were structurally sound, the physics of the radiation domain and the physics of the matter domain would long ago have merged into a single theory. The historical record shows the opposite. Quantum field theory—the physics of radiation, of massless gauge fields [13], of the null domain—and general relativity—the physics of matter, of mass-energy curving the E4D manifold [14]—have resisted unification for one hundred years, producing non-renormalisable infinities at every attempted junction [15,16].

The present framework does not rest this section on the historical record alone; the record is the symptom, and the disease is now stated formally. Any single-manifold unification U4D must contain three sectors and their couplings:

$$U4D = \{ T_{\text{massive}}, T_{\text{radiative}}, T_{\text{gravity}} \}$$

with the massive sector characterised by $m > 0$, $d\tau > 0$; the radiation sector by $m = 0$, $d\tau = 0$; and a dynamical interaction $T_{\text{massive}} \leftrightarrow T_{\text{radiative}}$. For the manifold to constitute one physical domain rather than a representational sheet over two, the framework must contain a continuous, Lorentz-invariant mapping

$$f: D_M \rightarrow D_R$$

connecting the sectors as states of one underlying temporal ontology. Sections V.1 and V.2 have proved that no such mapping exists: the Lorentz group contains no element joining the sectors (V.1),

and no finite-energy dynamical process performs the passage (V.2), because $\lim(v \rightarrow c) \gamma = \infty$ and the invariant momentum classes are disjoint. Formally:

$\neg \exists f : \tau_M \rightarrow \tau_R$ such that f is continuous and Lorentz invariant, while $\Delta\tau = \tau_M - \tau_R > 0$ invariantly.

The single-manifold assumption therefore requires a transformation that the Lorentz structure itself forbids. This is the precise sense in which the unification programme's difficulty is structural: the quantum-gravity impasse is a consistency test, and its persistent outcome is consistent with the two-domain separation and inconsistent with the single-domain reading. A successful unification theory should not be expected to merge the sectors into a single four-dimensional dynamical manifold; it should derive their coupling as a projection relation between domains—a prediction this framework states and Section VI supplies.

The mainstream treats the impasse as an unsolved technical problem. The present framework identifies it as a diagnostic result: the two theories cannot be unified within one geometric container because they are the native physics of two different domains. The infinities that erupt at every attempted merger are the same infinity that erupts in the Lorentz factor at $v = c$ —the mathematical protest of a formalism forced to treat a domain boundary as an interior point. The unification programme has not failed for lack of ingenuity. It has failed because its goal, as formulated, is a category error: it attempts to place radiation and matter— $\tau = 0$ and $\tau > 0$, the absolute and the relative, the massless and the massive—inside a single box whose own mathematics ruptures at exactly the seam where the two are joined.

VI. The Two-Domain Structure and the Source Geometry

The derivation is now complete in its negative half: the single-manifold picture is fractured at $v = c$ by coordinate-independent invariants, sealed by the absence of any connecting element in the Lorentz group, dynamically sealed by an infinite barrier, and empirically discredited by a century of failed unification. The positive half follows by asking what geometry the two domains, once separated, jointly imply.

The matter domain (E4D) does not generate the constant c ; it conforms to it. The radiation domain does not merely coexist with the matter domain; radiation arrives into the matter domain—it is emitted, it traverses, it is absorbed, it delivers energy (the photoelectric effect) [17], it delivers structure (all astronomical information), and its own state throughout is $\tau = 0$, meaning that from the radiation domain's side the delivery is instantaneous and processless. A domain whose contents appear within another domain instantaneously, carrying energy and information, while the receiving domain's geometry deforms to accommodate their invariant speed, stands to the receiving domain as a source stands to a projection surface. The geometric figure is not two adjacent boxes. It is a radiating source

and an illuminated interface: a spherical lamp, and the screen its light strikes. Matter's spacetime is the screen. Radiation is the light in transit. The source is the lamp.

The relation admits operator form, and the form predicted at the close of Section V.3 is exactly this. The coupling of the domains is a projection

$$P : S^6D \rightarrow S^3D, O_matter = P(O_source),$$

where P is non-invertible: $P^{-1}(O_matter)$ does not exist as a physical operation, which is the formal statement that the matter domain cannot recover the full atemporal source state from within itself—the same irreversibility that the matter domain registers as the thermodynamic arrow and the information loss of every measurement.

Before the source geometry is named, one structural distinction must be recorded: Sections II–V establish the necessity of a separation between the radiation and matter temporal domains from Lorentz-invariant classification alone. They do not, however, uniquely determine the internal dimensional architecture of either domain. The six-dimensional realisation presented below is therefore not a prerequisite for the derivation of the two-domain separation, but a constructive extension that specifies one possible source geometry and internal organisation of the already established domains.

This paper names the source S^6D , adopting the notation and the geometry of the Ripple-Instantiation cosmogenesis model [18], which independently derived—from the JWST high-redshift luminosity anomalies [19,20]—a six-layer spherical cascade: S^6D , a bounded spherical nucleus at absolute rest ($T = 0$), projecting instantaneously outward through S^5D (the atemporal configuration manifold) and S^4D (the quantum-field conversion interface) into S^3D (the observable material universe), with S^2D and S^1D as sub-material layers. The convergence of the two derivations is the central result of this paper. The Ripple model reached the six-layer structure from cosmological observation downward. The present argument reaches the same bifurcation from the foundations of mechanics upward: the radiation domain of this paper is the upper cascade S^6D – S^5D – S^4D , whose native state is $T = 0$ —exactly the proper time of every photon; the matter domain of this paper is the lower cascade S^3D – S^2D – S^1D , whose native state is emergent ordered time $\tau > 0$ —exactly the proper time of every massive body. Newton's absolute time is the state of the upper cascade. Newton's relative time is the emergent parameter of the lower cascade. Einstein's proper-time bifurcation is the seam between them, and the invariance of c is the seam made measurable. The detailed internal structure of the two triads is established in the Ripple-Instantiation framework (part of the SDMC dimensional theory), and is not re-derived here.

VII. Consequences and Resolved Anomalies

The two-domain structure dissolves, rather than solves, a set of standing paradoxes, because each of them is generated by the single-manifold assumption and by nothing else.

The photon rest-frame paradox: relativity forbids a photon rest frame, yet places photons inside a manifold defined by frames. In the two-domain structure the prohibition is explained: the photon has

no rest frame in E4D because the photon is not a resident of E4D; one cannot occupy a frame in a domain to which one does not belong. In the vocabulary of Section IV.1: the photon is geometrically included in M_4 and ontologically excluded from D_M , and the missing rest frame is the exact point where the two relations come apart.

The twin paradox: the asymmetry between the travelling and staying twin, officially attributed to acceleration, quietly presupposes a background against which acceleration is absolute. The two-domain structure supplies that background legitimately: the radiation domain is the absolute reference that the single-manifold doctrine uses in practice while denying in principle.

Quantum non-locality: entanglement correlations [21] that are instantaneous in E4D coordinates are paradoxical only if the correlated pair [22] is fundamentally an E4D object. If the pair is a single structure in the atemporal upper cascade (where $T = 0$ and no signal "travels" because no duration exists), its two S^3D appearances correlate without transmission—the correlation is projective, not causal, and the prohibition on superluminal signalling within E4D is untouched. Formally, the pair is one O_source with two images under P ; correlation between the images requires no dynamics between them.

The quantum–gravity impasse: as argued in Section V.3, the impasse is reclassified from an unsolved problem to a confirmed prediction: theories native to different domains do not unify, and the framework asserts they never will—a claim that is itself falsifiable (Section VIII). The constructive corollary stands with it: the correct junction of the two physics is the projection relation P , not a merged manifold.

VIII. Falsification Conditions

The framework is asserted as physical truth and therefore states the conditions under which it is false.

(1) If any massive particle is physically accelerated to $v = c$ with finite energy, the infinite-energy barrier of Section V.2 is void and the two-domain separation collapses.

(2) If a quantum theory of gravity is completed that renormalises consistently within a single four-dimensional manifold, without invoking additional dimensional structure, holographic projection, or an atemporal substrate [23], then Section V.3's diagnostic is refuted.

(3) If the invariance of c is derived as an internal theorem of matter-domain dynamics—that is, if a frame-dependent mechanism is shown to generate frame-independence—then Section IV's signature argument fails.

(4) If a coordinate system is exhibited in which the proper time of a null worldline is non-zero, Section V.1 fails; this condition is stated for completeness, its impossibility being a theorem of Lorentzian geometry—which is precisely the point: the fracture is invariant.

(5) If a Lorentz transformation, or any continuous Lorentz-invariant mapping $f : D_M \rightarrow D_R$, is constructed, the group-theoretical separation of Sections V.1 and V.3 fails; its non-existence is likewise a theorem of the Lorentz structure, and the condition is stated to make the framework's dependence on that structure explicit.

(6) The associated Ripple-Instantiation model carries its own observational falsifiers (the super-horizon correlation floor $\xi_{\text{res}}(r)$, the cross-redshift coherence floor Σ_0 , and the persistent CMB–LSS alignment) [24]; failure of all three under JWST and Euclid data falsifies the specific six-layer realisation of the source geometry, though not the two-domain derivation itself, which stands on Sections II–V independently.

IX. Conclusion

Newton defined absolute time and assigned every clock to relative time. Einstein proved that light's proper time is zero and matter's is positive, and axiomatised a speed that no frame can alter. Assembled without addition, these two bricks build one structure: the universe contains a radiation domain in the state of absolute time and a matter domain in the state of relative time; the boundary between them is invariant, impassable, absent from the theory's own symmetry group, and marked in every laboratory by the constancy of c ; the single-manifold doctrine that denies the boundary ruptures at it mathematically, dynamically, and historically. The two domains resolve into the upper and lower triads of a six-dimensional spherical cascade whose source, S^6D , radiates the projection that matter inhabits, coupled to it by a non-invertible projection P . Two hundred years of Newton and one hundred and twenty years of Einstein were both correct—about different domains. The error was the box.

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